# Data Analytics day wise material

# Lecture–1: Introduction to Data Analytics (DA)

**Topics:**

1. Motivation to Study DA
2. Syllabus Discussion
3. Roadmap to Placement
4. Roles to Apply for Placement
5. Installation of Anaconda & Required Libraries

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# 1. Motivation to Study Data Analytics

## What is Data Analytics?

Data Analytics is the process of:

**Collecting → Cleaning → Transforming → Analyzing → Visualizing → Finding Insights → Decision Making**

Simple example:

Suppose Zomato notices food orders decreasing.

Questions:

- Which city has fewer orders?
- Which food category dropped?
- Which age group stopped ordering?
- Does rainy weather affect orders?

Analyzing data to answer these questions = Data Analytics.

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## Why Study Data Analytics?

### Reason 1: High Demand Jobs

Companies hiring:

- Amazon
- Flipkart
- Google
- TCS
- Infosys
- Accenture
- Deloitte
- Microsoft
- Swiggy
- Zomato

Need:

- Data Analysts
- BI Analysts
- Product Analysts
- Reporting Analysts

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## Reason 2: Good Salary

Approximate India salaries:

Freshers:

₹4–8 LPA

1–3 years:

₹8–15 LPA

Experienced:

₹15–30+ LPA

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## Reason 3: Non-CS students can enter

Students from:

- BCA
- BTech
- BSc
- BCom
- MBA
- MCom
- Mechanical
- Civil

all can learn DA.

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### Reason 4: Easier than Data Science

Data Science:

Programming + ML + Deep Learning

Data Analytics:

Excel + SQL + Python + Power BI + Statistics

Hence easier for beginners.

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### Reason 5: Every industry uses analytics

Examples:

Healthcare:
Predict diseases

Banking:
Fraud detection

Education:
Student performance

E-commerce:
Customer behavior

Sports:
Player performance

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# Data Analytics Workflow

Think of making tea ☕

Raw milk → boil → add sugar → filter → serve

Similarly:

Raw Data

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Clean Data

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Analyze Data

↓

Visualize Data

↓

Business Decision

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# 2. Syllabus Discussion

Your syllabus contains:

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## Module 1: Python Libraries

### Pandas

Used for:

- Reading CSV
- Cleaning data
- Filtering
- Grouping

Example:

Customer dataset analysis

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### NumPy

Used for:

Fast numerical operations

Example:

Matrix calculations

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### Matplotlib

Used for:

Graphs

Examples:

Bar chart
Pie chart
Histogram

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### Seaborn

Advanced visualization

Example:

Heatmap

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### Plotly

Interactive graphs

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# Module 2: EDA (Exploratory Data Analysis)

Topics:

- Missing values
- Duplicate values
- Outliers
- Cleaning

Real-world:

Messy company datasets

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# Module 3: Statistics

Need:

- Mean
- Median
- Probability
- Hypothesis testing
- A/B testing

Interview favorite section.

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# Module 4: Excel

Need:

- Pivot table
- VLOOKUP
- XLOOKUP
- Macros

Asked in analyst interviews.

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# Module 5: Power BI

Need:

Dashboard creation

Example:

Sales dashboard

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# Module 6: DAX

Need:

Power BI calculations

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# Module 7: Projects

Need:

Real datasets

Portfolio creation

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# Skills after completing syllabus

You should know:

✓ Python
✓ Pandas
✓ Statistics
✓ Excel
✓ Power BI
✓ DAX
✓ EDA

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# Missing Important Placement Topics

Need extra learning:

1. SQL (VERY IMPORTANT)
2. Git/GitHub
3. ML Basics
4. Feature Engineering
5. Dashboard Storytelling
6. APIs
7. GenAI

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# 3. Roadmap to Placement

Recommended roadmap:

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# Stage 1 (Month 1)

Learn:

- Python basics
- Pandas
- NumPy

Practice:

Small datasets

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# Stage 2 (Month 2)

Learn:

- Visualization
- EDA

Practice:

Projects

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# Stage 3 (Month 3)

Learn:

Statistics

Topics:

Mean, Median, Hypothesis testing

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# Stage 4 (Month 4)

Learn:

Excel + Power BI

Create dashboards

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## Stage 5 (Month 5)

Learn:

SQL

(Most important for placements)

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## Stage 6 (Month 6)

Projects + Resume + Interview prep

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Roadmap:

Python

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Pandas + NumPy

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EDA + Statistics

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Excel + Power BI

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SQL

↓

Projects

↓

Resume

↓

Placement

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# 4. Roles to Apply for Placement

After course completion:

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## Role 1: Data Analyst

Work:

Analyze business data

Skills:

Python
Excel
SQL
Power BI

Salary:

4–10 LPA

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## Role 2: Business Intelligence Analyst

Work:

Dashboard creation

Need:

Power BI

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## Role 3: Reporting Analyst

Work:

Generate reports

Need:

Excel + SQL

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## Role 4: Product Analyst

Work:

User behavior analysis

Need:

Statistics + SQL

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## Role 5: Business Analyst

Need:

Communication + Analytics

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## Role 6: Junior Data Scientist

Need:

ML + Statistics

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## Role 7: Operations Analyst

Need:

Excel + Dashboard

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# Placement Priority

Apply in order:

1. Data Analyst ⭐⭐⭐⭐⭐
2. BI Analyst ⭐⭐⭐⭐⭐
3. Reporting Analyst ⭐⭐⭐⭐
4. Product Analyst ⭐⭐⭐⭐
5. Business Analyst ⭐⭐⭐

6. Junior Data Scientist ⭐⭐⭐

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# 5. Installation of Anaconda & Required Libraries

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## What is Anaconda?

Anaconda is a software package containing:

Python + Libraries + Jupyter Notebook

One installation gives:

- Python
- Pandas
- NumPy
- Matplotlib
- Jupyter

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## Step 1: Download Anaconda

Go to:

Anaconda Official Website

Download:

Windows → Python 3.x version

Install normally:

Next → Next → Finish

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## Step 2: Open Anaconda Navigator

Search:

Anaconda Navigator

Open it

You will see:

- Jupyter Notebook
- Spyder
- VS Code

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## Step 3: Launch Jupyter Notebook

Click:

Launch

Browser opens

Create:

New → Python Notebook

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## Step 4: Test Python

Write:

```
print("Hello Data Analytics")
```

Output:

```
Hello Data Analytics
```

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# Required Libraries

Install:

```
pip install pandas
pip install numpy
pip install matplotlib
pip install seaborn
pip install plotly
pip install openpyxl
pip install scikit-learn
pip install scipy
```

Install all together:

```
pip install pandas numpy matplotlib seaborn plotly openpyxl scikit-learn
scipy
```

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# Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

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Test:

```
import pandas as pd

print(pd.__version__)
```

If version prints → installation successful.

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# Lecture-1 Summary

After this lecture students should know:

✓ What is Data Analytics
✓ Why DA is important
✓ Career opportunities
✓ Placement roadmap
✓ Job roles
✓ Complete syllabus overview
✓ Install Anaconda
✓ Install libraries
✓ Run first Python code